The proposition 1 tells us that the kernel sequence $\{\rho_\ell\}$ is precisely given by $\rho, \kappa(\rho), \kappa(\kappa(\rho)), \dots$. Thus, we can analyze its convergence by studying the fixed points of the kernel map κ, which are the values of ρ^* that satisfy $\kappa(\rho^*) = \rho^*$. With the assumption that $\mathbb{E}\,\phi(X)^2 = 1$, the kernel map κ is a mapping between $[-1, -1]$ to itself. Thus, Brower's fixed point theorem implies that the kernel map κ has at least one fixed point ρ^*. However, as we will show, there is potentially more than one fixed point, and it will be interesting to understand which ones are locally or globally attractive.

5 Hermite expansion of activation functions

Hermite polynomials possess completeness and orthogonality under the Gaussian measure. Therefore, any function that is square-integrable with respect to a Gaussian measure can be expressed as a linear combination of Hermite polynomials (see below). The square integrability rules out the possibility of having heavy-tailed postactivations without second moments. This holds for all activations that are used in practice. We use *normalized* Hermite polynomials and their coefficients.

Definition 2. Normalized Hermite polynomials $\mathrm{he}_k(x)$ are defined as follows

$$\mathrm{he}_k(x) := \frac{1}{\sqrt{k!}}(-1)^k e^{\frac{x^2}{2}} \frac{d^k}{dx^k} e^{-\frac{x^2}{2}}.$$

Although Hermite polynomials have been used in probability theory and statistics, particularly in the context of Gaussian processes [Williams and Rasmussen, 2006], their application in analyzing neural network dynamics provides a novel methodological tool. Previous works, such as Daniely et al. [2016], have utilized polynomial expansions to study neural networks, but our explicit use of Hermite polynomial expansions to derive the kernel map and analyze convergence is a new contribution.

The crucial property of Hermite polynomials is their orthogonality:

$$\mathbb{E}\,\mathrm{he}_k(X)\mathrm{he}_l(X) = \delta_{kl}, \qquad X \sim \mathcal{N}(0,1). \tag{4}$$

Scaling by $1/\sqrt{k!}$ ensures that polynomials form a *orthonormal* basis. Based on this property, we can define:

Definition 3. Given a function ϕ, square-integrable with respect to the Gaussian measure. Its Hermite expansion is given by:

$$\phi = \sum_{k=0}^{\infty} c_k \mathrm{he}_k, \qquad c_k = \mathbb{E}\,\phi(X)\mathrm{he}_k(X), \qquad X \sim \mathcal{N}(0,1)\,.$$

Here, c_k are called the Hermite coefficients of ϕ.

In addition to orthogonality, Hermite polynomials have another "magical" property that is crucial for our later analysis.

Lemma 1 (Mehler's lemma)**.** *For standard-normal random variables X, Y with covariance ρ it holds*

$$\mathbb{E}\,\mathrm{he}_m(X)\mathrm{he}_n(Y) = \rho^n \delta_{mn}, \qquad \begin{pmatrix} X \\ Y \end{pmatrix} \sim \mathcal{N}\left(0, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}\right).$$

This lemma states that given two Gaussian random variables X, Y with covariance ρ, the expectation of the product of Hermite polynomials is zero unless the indices are equal (see explanations in Section A for proof). The lemma 1 is crucial for our theory. Based on this lemma, we can express the kernel map κ in terms of the Hermite coefficients, showing a particular structure of the kernel map.

Corollary 1. *We have the following explicit form for the kernel map:*

$$\kappa(\rho) = \sum_{k=0}^{\infty} c_k^2 \rho^k.$$

The proof follows directly from the definition of kernel map, expanding ϕ in the Hermite basis and then applying Lemma 1.

Daniely et al. [2016] show a similar analytic form for dual activation, which aligns with our definition of kernel map. However, they did not use this to study global dynamics. To our knowledge, the only study that uses Hermite polynomials to study global dynamics is [Joudaki et al., 2023b], which is limited to centered activations. i.e., those that satisfy $\mathbb{E}\,\phi(X) = 0$, or equivalently $\kappa(0) = c_0^2 = 0$.

6 Convergence of the kernel with general activations

In this section, we will show that for any nonlinear activation function, there is a unique fixed point ρ^* that is globally attractive.

Theorem 1. *Given a nonlinear activation function ϕ with kernel map κ such that $\kappa(1) = 1$. Define the kernel sequence $\rho_{\ell+1} = \kappa(\rho_\ell)$, with $\rho_0 \in (-1, 1)$. Then there is a unique globally contracting fixed point ρ^*, which is necessarily non-negative $\rho^* \in [0, 1]$. The only other fixed points distinct from ρ^*, could be ± 1, neither of which is stable. Furthermore, we have the following contraction rate towards ρ^*:*

1. *If $\kappa(0) = 0$ then $\rho^* = 0$ is an attracting with rate*

$$\frac{|\rho_\ell|}{1 - |\rho_\ell|} \leq \frac{|\rho_0|}{1 - |\rho_0|}\alpha^\ell, \qquad \alpha := \frac{1}{2 - \kappa'(0)}.$$

2. *If $\kappa(0) > 0$ and $\kappa'(1) < 1$, then $\rho^* = 1$ is attracting, with rate*

$$|\rho_\ell - 1| \leq |\rho_0 - 1|\alpha^\ell, \qquad \alpha := \kappa'(1).$$

3. *If $\kappa(0) > 0$, and $\kappa'(1) = 1$ then $\rho^* = 1$ is attracting with rate*

$$|\rho_\ell - 1| \leq \frac{|\rho_0 - 1|}{\ell\alpha|\rho_0 - 1| + 1}, \qquad \alpha = 1 - \kappa(0) - \kappa'(0).$$

4. *If $\kappa(0) > 0$, and $\kappa'(1) > 1$ then the attracting fixed point is necessarily in the range $\rho^* \in (0, 1)$, satisfying $\kappa'(\rho^*) < 1$, for which we have*

$$|\rho_\ell - \rho^*| \leq \frac{|\rho_0 - \rho^*|}{1 - |\rho_0|}\alpha^\ell \qquad \alpha = \max\left\{1 - \kappa(0), \kappa'(\rho^*), \frac{1 - \rho^*}{2 - \kappa'(\rho^*)}\right\},$$

where $\alpha < 1$.

Implications. Let us take a step back and review the main takeaway of Theorem 1. Omitting the constants and for sufficiently large depth ℓ, we have

$$\langle h^\ell(x), h^\ell(y)\rangle_{\text{avg}} = \begin{cases} 0 + O(\alpha^\ell \langle x, y\rangle_{\text{avg}}) & \text{case 1} \\ 1 + O(\alpha^\ell \langle x, y\rangle_{\text{avg}}) & \text{case 2} \\ 1 + O\left(\langle x, y\rangle_{\text{avg}}/\ell\alpha\right) & \text{case 3} \\ \rho^* + O(\alpha^\ell \langle x, y\rangle_{\text{avg}}) & \text{case 4,} \end{cases}$$

where $\langle x, y\rangle_{\text{avg}}$ denotes the input similarly.

Broadly speaking, we can think of three categories of bias:

- *Orthogonality bias* $\rho^* = 0$: Implies that as the network becomes deeper, representations are pushed towards orthogonality exponentially fast. We can think of this case as a bias towards independence.
- *Weak similarity bias* $\rho^* \in (0, 1)$: Implies that as the network becomes deeper, the representations form angles between 0 and $\pi/2$. Thus, in this case, the representations are neither completely aligned nor completely independent.
- *Strong similarity* $\rho^* = 1$: Implies that as the network becomes deeper, representations become more similar or aligned, as indicated by inner products converging to one, exponentially fast in case 2, polynomially in case 3.

See section B for a review of commonly used activations, and their biases according to Theorem 1.

The bias of activation and normalization layers has been extensively studied in the literature. For example, Daneshmand et al. [2021] show that batch normalization with linear activations makes representations more orthogonal, relying on a technical assumption that is left unproven (see assumption $\mathcal{A}_1$). Similarly, Joudaki et al. [2023a] extend this to odd activations yet introduce another technical assumption about the ergodicity of the chain, which is hard to verify or prove (see Assumption 1). In a similar vein, Meterez et al. [2024] show the global convergence towards orthogonality in a network with batch normalization and orthogonal random weights, but do not theoretically analyze nonlinear activations. Yang et al. [2019] prove the global convergence of the Gram matrix of a network with linear activation and batch

normalization (see Corollary F.3) toward a fixed point. However, the authors explain that because they cannot establish such a global convergence for general activations (see page 20, under the paragraph titled Main Technical Results), they resort to finding locally stable fixed points, meaning that if preactivations have such a Gram matrix structure and they are perturbed infinitesimally. Recent studies have also examined the evolution of covariance structures in deep and wide networks using stochastic differential equations [Li et al., 2022], which is similar to kernel dynamics and kernel ODE introduced here. However, the covariance SDE approach does not theoretically show the global convergence towards these solutions. Finally, Joudaki et al. [2023b] establish a global bias towards orthogonality for centered activations (see Theorem A2), which aligns with case 1 of Theorem 1, and to extend to other activations, they add layer normalization after the activation to make it centered. In contrast, Theorem 1 covers all existing activations.

One alternative way of interpreting the results of Theorem 1 is that as we pass two inputs through nonlinear activations, it 'forgets' the similarity between inputs and converges to a fixed point value that is independent of the inputs and only depends on the activation. Taking the view that the network ought to remember some similarity of the input for training, we can leverage Theorem 1 to strike a balance between depth and nonlinearity of the activations, as argued in the following.

Corollary 2. *Let $\epsilon > 0$ be the some numerical precision, and let ϕ be an activation function such that $\mathbb{E}\,\phi(X)^2 = 1$, with its kernel map obeying $\kappa'(1) < 1$. Then, for any initial pair of inputs x and y after $\ell \geq L := \frac{\ln(1/\epsilon)}{\ln(1/\kappa'(1))}$ layers, the representations of x and y will become numerically indistinguishable.*

Approximately, $\ln(1/\epsilon)$ represents the number of bits of numerical precision available in floating-point representations. For example for `float32` numbers, $\ln(1/\epsilon)$ is approximately $\ln(1/2^{-128}) \approx 88$. Therefore, when the network depth ℓ exceeds threshold $88/\ln(1/\kappa'(1))$, the representations of any two inputs will become numerically indistinguishable from one another. This effect propagates through subsequent layers, potentially making training ineffective or impossible. Notably, ReLU and sigmoid satisfy the conditions of this corollary, implying that this phenomenon can occur in networks utilizing these activations. Previous works on rank collapse [Daneshmand et al., 2020, Noci et al., 2022] and covariance degeneracy [Li et al., 2022] have observed and studied this phenomenon in various settings. However, the explicit and closed-form relationship presented in Corollary 2 provides a novel and precise quantification of this effect.

One of the most striking results of this theorem is that for any nonlinear activation, there is exactly one fixed point ρ^* that is globally attractive, while other fixed points are not stable. Furthermore, the fixed point is necessarily non-negative. This implies that for any MLP with nonlinear activations, the covariance of preactivations will converge towards a fixed point, which is always non-negative. It turns out that there is a geometric reason why no globally attracting negative fixed point could exist (See remark S.2)

Theorem 1 has several aspects that may seem counterintuitive. For instance, case 1 demonstrates that a quantity $|\rho|/(1-|\rho|)$ decays exponentially as depth increases. To gain a better understanding of the theories behind Theorem 1, we can cast our layer-wise kernel dynamics as a continuous-time dynamical system, which is discussed next. For a detailed proof of Theorem 1, please refer to Section A.

6.1 A continuous time differential equation view to the kernel dynamics

One of the most helpful ways to find insights into discrete fixed point iterations is to cast them as a continuous problem. More specifically, consider our fixed point iteration:

$$\Delta\rho_\ell = \rho_{\ell+1} - \rho_\ell = -\rho_\ell + \sum_{k=0}^{\infty} c_k^2 \rho_\ell^k.$$

We can replace this discrete iteration with a continuous time differential equation, which we will refer to as the kernel ODE:

$$d\rho/dt = -\rho + \sum_{k=0}^{\infty} c_k^2 \rho^k. \qquad \text{(kernel ODE)}$$

Recent studies have introduced stochastic models such as the Neural Covariance SDE by Li et al. [2022], which can be viewed as the stochastic analog of kernel ODE. However, kernel ODE will capture the most important parts of the discrete kernel dynamics for our main purpose of characterizing the attracting fixed points. The continuous analog of fixed points is a ρ^* that satisfies $\rho'(\rho^*) = 0$. Furthermore, the fixed point is globally attracting if we have ρ' become negative for $\rho > \rho^*$ and positive for $\rho < \rho^*$. We can say that the fixed point is locally attractive if this property only holds in some small neighborhood of ρ^*.

While the kernel ODE presents an interesting transformation of the problem, it does not necessarily have a closed-form solution in its general form. However, our main strategy is to find worst-case (slowest) scenarios for convergence that